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SETP4583- SURFACTANT AND DETERGENT TECHNOLOGY

INDIVIDUAL REFLECTIVE JOURNAL

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Figure 1: Our class photo of course Surfactant and Detergent Technology

Introduction

When the semester started, I considered the Surfactant and Detergent Technology course to be a topic that was primarily centered on chemical equations, raw materials, and production. Initially, I was truly puzzled by the amount of surfactants present and the numerous new terms used in the introductory lectures. My initial thought of the course was that it would be very technical and theoretical with the main focus on the calculations and definitions of chemistry. Back then, I was not aware of the extent to which the science of detergent is related to the daily activities and real-life products that individuals encounter in their everyday life.

I decided to enroll in this course since the title of the course is interesting. I had an initial objective of just passing the course with a good grade and having a basic idea of the surfactant. But as the weeks went by, my objectives gradually changed. I did not think that even a basic lab exercise would alter my perception of ordinary items in the home that I had always assumed.

Reflection on Key Learnings

The behavior and role of surfactants at the molecular level was one of the most significant concepts that I obtained during this course. The information on hydrophilic heads and hydrophobic tails made me realize the way the cleaning process works, not the definitions I remembered by heart. This was further understood in the laboratory sessions when I was able

to directly see the interaction of surfactants with oil and water to create emulsions. And that was literally the first day I got into a lab and work as opposed to staring at the computer monitor. It was one of my initial “aha” moments when theory that was taught in lectures became practical.

The lectures gave a good background in various forms of surfactants, including anionic, non-ionic, and amphoteric surfactants. In the beginning, it was difficult to recall the functions and features of each type. Nevertheless, after several exposures in laboratories and during group discussions, I started to see the reason behind the selection of some particular surfactants in particular products. To take an example, I got to know the reason why anionic surfactants such as SLES and LABS are useful in removing grease, yet when not combined with milder surfactants such as betaine, they can be irritating.

The role of laboratory sessions was rather significant in strengthening my knowledge. In contrast to the completely theoretical topics, this course provided me with an opportunity to see with my own eyes the results of such processes as the formation of foam, emulsification, and texture modifications, which made the learning process more exciting and memorable. Simple experiments like foaming ability observation assisted me in relating abstract concepts to practical outcomes. I also understood that even minor changes in formulation could have a great impact on the performance of products. My learning was further reinforced through the group project in which we came up with a prototype of a dishwashing detergent. It was the first occasion when I felt that I was engaged in an activity close to real industrial work and not a routine university lab exercise. Teamwork enabled me to hear other points of view, share ideas and the value of teamwork in product development.

The other important lesson was that the formulation of detergents does not just revolve around the cleaning efficiency. In the discussions and case examples, I understood that the failure of products in the market can be frequently associated with the inadequate attention to the comfort of users, their safety, or the environmental impact, but not cleaning power. The factors include skin-friendliness, environmental impact, cost, and consumer expectations which are also essential.



Figure 2: Prototype dishwashing detergent developed during the group project

Application and Real-Life Relevance

As the course progressed, I started to realize the obvious links between what we studied and what happens in the industry. Different products such as dishwashing liquids, shampoos, and car cleaners are not mere items on shelves in the stores to me. Now I understand why they chose to formulate some products more or less foaming, or why some of them feel less harsh on the skin.

The group project was particularly useful in demonstrating how theoretical knowledge can be used in real life situations. The use of limited equipment was also realistic challenges encountered in the early stage of product development. When we created our own detergent, I gained first-hand experience on the need to balance on performance, safety, and cost by companies. As an example, excess anionic surfactant can enhance cleaning efficacy, but can lead to skin irritation, whereas insufficiently can lead to reduced efficacy. This kind of decision making is evidently applicable in the development of industrial products.

I also learnt about sustainability and environmental issues in the detergent industry. The information about biodegradability and toxicity helped me understand that it is the duty of manufacturers to be responsible to the environment. This is specifically applicable in the modern world where consumers are increasingly becoming eco-friendly. I can envision using

this knowledge in my future jobs in the field of quality control, research and development or even regulatory compliance.

Challenges and Growth

Learning the chemical terms and processes that take place in the behavior of surfactants was also among the primary challenges that I encountered in this course. Sometimes, I was discouraged whenever I was not able to learn some concepts instantly, particularly in the first few weeks of the semester. The initial material was initially daunting, particularly when several types of surfactants and their relationships were presented simultaneously. I was not always able to relate equations and molecular forms to real-life results.

To eliminate this, I took extra time going through lecture notes and talking to my classmates on concepts. Furthermore, being an active participant in the lab sessions also contributed to my learning in a more effective way because it was easier to observe something and comprehend the complex ideas. The second issue was collaborating with a group of people since the team members had various working styles and views. Nevertheless, this experience helped me to learn how to communicate more effectively, compromise when it is needed, and value collaboration.

These struggles helped me to become more tolerant of myself and more willing to ask questions. I also felt more confident in how I can use theoretical knowledge in practical tasks and I believe that this is a great personal development.



Figure 3: Group photo G1

Personal and Professional Development

This course has positively influenced my mindset toward applied science and product development. I now have a deeper appreciation for how chemistry impacts daily life and consumer products.

Professionally, this course has sparked my interest in formulation science and industrial applications of chemistry. It has also increased my awareness of career pathways related to quality control and sustainable product innovation. In the future, I would like to explore topics like sustainable surfactant development. Overall, this course has not only enhanced my academic knowledge but also helped me develop a more practical and industry-oriented perspective that will be valuable in my future career.